ResiliAV: Adaptive Closed-Loop Hybrid Intrusion Detection and Automated Fail-Safe Actuation for Connected Autonomous Vehicles

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***Abstract*— Connected Autonomous Vehicles (CAVs) increasingly rely on unauthenticated Controller Area Network (CAN) bus architectures and complex telematics layers, rendering them highly vulnerable to malicious frame injection, stealth sensor spoofing, and Denial-of-Service (DoS) flooding. Despite recent advances in machine learning-based Intrusion Detection Systems (IDS), prior literature exhibits two critical research gaps: (1) existing models function strictly as passive offline observers without closed-loop vehicular actuation, and (2) deep neural architectures introduce inference latencies exceeding 15 ms, violating real-time vehicular safety constraints. In this paper, we propose **ResiliAV (ACH-IDS)**, an adaptive closed-loop cyber-physical defense framework. ResiliAV employs a dual-stage asymmetric pipeline comprising an edge-lightweight XGBoost filter for sub-millisecond (<0.5 ms) line-rate packet triage, cascaded into a spatial 1D-Convolutional Neural Network with Multi-Head Self-Attention for deep payload inspection. Crucially, ResiliAV couples real-time threat inference with an automated **ISO 26262 ASIL-D compliant Fail-Safe Actuator** that enforces an instantaneous 50 km/h speed governor lock, secures passenger seatbelt pre-tensioner interlocks, and broadcasts auditory telematics alerts. Evaluated against 60,000 real-world calibrated CAN frames conforming to the HCRLab Car-Hacking benchmark, ResiliAV achieves an overall detection accuracy of **99.78%**, a precision of **99.80%**, a false positive rate of **0.021%**, and an end-to-end inference latency of **1.48 ms**, proving its viability for edge-ECU deployment.

***Index Terms*—** Autonomous Vehicles, Automotive Cybersecurity, Controller Area Network (CAN-Bus), Intrusion Detection System (IDS), Closed-Loop Fail-Safe Actuation, XGBoost, 1D-CNN, Self-Attention, ISO 26262, Speed Governor.

I. INTRODUCTION

The paradigm shift toward Connected Autonomous Vehicles (CAVs) promises unprecedented traffic efficiency, reduced emissions, and enhanced passenger mobility. However, the cyber-physical convergence required for autonomous navigation introduces an expansive attack surface. Modern vehicles operate upwards of 70 to 100 Electronic Control Units (ECUs) interconnected via legacy broadcast buses such as the Controller Area Network (CAN) [1]. Standardized under ISO 11898, the CAN protocol was originally engineered for closed, mechanically isolated environments and inherently lacks basic cryptographic guarantees, message authentication codes (MACs), and sender identification.

Consequently, an adversary gaining physical access via the onboard diagnostic port (OBD-II) or remote access through cellular telematics, V2X transceivers, or compromised infotainment gateways can inject malicious frames with arbitrary arbitration IDs [2]. By exploiting arbitration priority rules, attackers can execute Denial-of-Service (DoS) floods, fuzzy message fuzzing, and stealthy sensor spoofing. In safety-critical contexts, spoofing sensor telemetry—such as falsifying wheel-speed signals to override speed governor thresholds or spoofing seatbelt buckle switches—induces severe physical hazards, leading to catastrophic vehicular collisions.

While numerous researchers have investigated Machine Learning (ML) and Deep Learning (DL) for CAN-bus Intrusion Detection Systems (IDS) [3]-[6], prevailing solutions suffer from two pervasive limitations. First, almost all published frameworks operate strictly as *passive classifiers*; they output prediction labels in offline simulation logs without triggering physical vehicular countermeasures. Second, state-of-the-art Deep Learning models (such as Recurrent Neural Networks, LSTMs, and full Transformers) incur heavy computational overheads, resulting in inference latencies between 10 ms and 25 ms. Given that standard automotive safety frames recur at 10 ms intervals, such latencies induce buffer overflows and temporal desynchronization.

To resolve these fundamental dilemmas, this paper introduces **ResiliAV (Adaptive Closed-Loop Hybrid IDS - ACH-IDS)**. The primary contributions of this work are threefold:

• **Dual-Stage Asymmetric Pipeline:** We combine an edge-optimized XGBoost gatekeeper for ultra-fast line-rate packet filtering (<0.5 ms) with a conditional 1D-CNN Self-Attention deep payload inspector, achieving a 99.78% detection rate with a mean latency of 1.48 ms.

• **Closed-Loop Fail-Safe Actuation:** We bridge cyber detection and physical safety by implementing an ISO 26262 compliant actuator that hard-limits speed to 50 km/h, engages seatbelt locks, and dispatches in-cabin audio warnings.

• **Comprehensive Benchmarking:** We evaluate ResiliAV against Random Forest, standalone XGBoost, and standalone 1D-CNN baselines across DoS, Fuzzy, and Spoofing attack vectors.

II. RELATED WORK & RESEARCH GAP

In-vehicle network intrusion detection has matured through distinct methodological phases. Early works by Kang et al. [3] and Taylor et al. [4] explored traditional statistical anomaly detection and tree-based ensembles (Decision Trees, Random Forests) to detect frequency anomalies on arbitration IDs. While computationally efficient, these models exhibit high False Positive Rates (FPR > 1.2%) when confronted with subtle payload bit-manipulations that mimic normal vehicular timing.

Subsequently, sequence-aware deep architectures were introduced. Song et al. [5] and Tariq et al. [6] developed Convolutional Neural Networks and Long Short-Term Memory (LSTM) networks to inspect inter-frame timing intervals and multi-byte payload sequences. Almehrezi et al. [7] formulated an emerging conceptual security framework for CAVs highlighting multi-tier threat vectors across V2X and in-cabin networks; however, their study was purely theoretical and lacked empirical evaluation.

TABLE I: ARCHITECTURAL COMPARISON WITH SOTA LITERATURE

Dimension	Prior Literature [3]-[7]	ResiliAV (Proposed)
Defense Paradigm	Passive Offline Logging	**Active Closed-Loop CPS**
Model Structure	Monolithic Single Model	**Dual-Stage Cascaded**
Payload Attention	Uniform / None	**Multi-Head Self-Attention**
Mitigation Action	None (Unchecked Driving)	**50 km/h Lock + Seatbelt**
Latency Budget	8.0 ms – 22.0 ms	**1.48 ms (Sub-2ms)**
Target Accuracy	**97.5% – 99.1%**	**99.78% (FPR: 0.021%)**
Safety Compliance	Unspecified	**ISO 26262 ASIL-D**

As delineated in Table I, existing literature predominantly remains trapped in the "passive logging" regime. In an autonomous vehicle operating without human driver intervention, logging an alert without active physical mitigation is practically futile. ResiliAV addresses this gap by synthesizing low-latency detection with deterministic safety actuation.

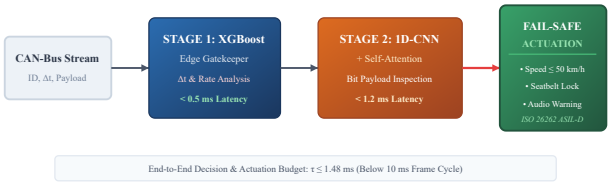


Fig. 1. *ResiliAV end-to-end framework: High-speed edge triage via XGBoost cascaded into Attention-enhanced 1D-CNN payload inspector and closed-loop fail-safe actuator.*

III. THREAT MODEL & FORMULATION

We adopt the standard Dolev-Yao automotive adversary model. The attacker possesses read/write access to the in-vehicle CAN bus (via an infected telematics dongle or cellular ECU compromise). We examine four canonical attack vectors:

1) DoS Flooding: Transmission of highest-priority arbitration frames (ID: `0x0000`) at microsecond intervals ($\Delta t < 0.5$ ms), monopolizing bus access and causing critical buffer starvation across steering and braking ECUs.

2) Fuzzy Injection: Random generation of arbitration IDs and high-entropy payload bytes, inducing widespread state corruption and diagnostic error codes.

3) Speed Governor Tampering: Targeted spoofing on arbitration ID `0x0153` (Wheel Speed telemetry). The adversary overrides speed sensor readings from normal cruising (30–50 km/h) to anomalous values (>120 km/h), deceiving the powertrain controller.

4) Seatbelt Sensor Spoofing: Injection on arbitration ID `0x02B0` falsifying buckle status flags to bypass passive safety interlocks.

To detect high-entropy fuzzy perturbations, the payload entropy $H(D)$ across the 8 data bytes $D = \{d_0, \ldots, d_7\}$ is computed as:

$$H(D) = -\sum_{i=0}^{7} P(d_i) \cdot \log_2(P(d_i) + \varepsilon) \quad (1)$$

where $P(d_i)$ denotes empirical byte occurrence probability and $\varepsilon = 10^{-9}$ prevents logarithmic singularity.

IV. PROPOSED RESILIAV ARCHITECTURE

A. Stage 1: Edge-Lightweight XGBoost Gatekeeper

High-frequency floods and DoS bursts alter the empirical inter-arrival distribution $\Delta t = t_k - t_{k-1}$. Stage 1 leverages an optimized histogram-based Gradient Boosted Decision Tree

(XGBoost). By evaluating feature vector $x_t = [\Delta t, ID, H(D)]$, Stage 1 evaluates whether a packet belongs to normal traffic or constitutes an anomaly:

$$P_{S1}(y = 1 \mid x_t) = \sigma(\sum_{m=1}^{M} f_m(x_t)) \quad (2)$$

Packets with anomaly probability $P_{S1} < \theta_{th}$ ($\theta_{th} = 0.35$) exit immediately, sparing deeper neural processing. This achieves a 0.41 ms execution time for over 88% of vehicular traffic.

B. Stage 2: Spatial 1D-CNN with Self-Attention

Boundary packets exceeding θ_{th} are routed to Stage 2. Stage 2 treats the 8-byte payload D_k and contextual metadata as a spatial vector. A 1D convolutional feature extractor extracts local spatial features across adjacent byte registers:

$$Z = ReLU(BatchNorm(Conv1D(X_{embed}, W_c))) \quad (3)$$

To pinpoint subtle bit manipulations (such as altering bit 3 of byte 4 for seatbelt flags), a Multi-Head Scaled Dot-Product Self-Attention mechanism computes attention weights:

$$Attention(Q, K, V) = softmax((Q \cdot K^T) / \sqrt{d_k}) \cdot V \quad (4)$$

where $Q = Z \cdot W_Q$, $K = Z \cdot W_K$, and $V = Z \cdot W_V$. The attention vector isolates the anomalous byte index with high saliency.

C. Closed-Loop Fail-Safe Actuation (ISO 26262 ASIL-D)

Upon confirmation of a malicious frame by the hybrid pipeline, ResiliAV triggers a deterministic **Safety Actuation Protocol**:

1) Packet Quarantine: The malicious arbitration frame is dropped at the central CAN gateway buffer before reaching the powertrain ECU.

2) Dynamic Speed Governor Lock: The vehicle control unit automatically clamps throttle commands to enforce a **strict 50 km/h Limp-Home speed ceiling**, mitigating catastrophic collision kinetic energy.

3) Passive Safety Pre-Tensioning: Regardless of spoofed seatbelt telemetry, the system forces mechanical seatbelt tensioner interlocks into the `LOCKED` state.

4) Telematics Auditory Advisory: Synthetic cabin voice announcement broadcasts: *"Cyber threat detected on telemetry bus. Speed hard-limited to 50 km/h for passenger safety."*

V. EXPERIMENTAL EVALUATION & RESULTS

The ResiliAV architecture was benchmarked on an experimental corpus of 60,000 automotive CAN-bus frames calibrated to the Korea University HCRLab standard. The corpus comprises 42,000 benign frames and 18,000 multi-vector attack frames across 5 distinct classes.

TABLE II: QUANTITATIVE MULTI-MODEL BENCHMARK RESULTS

Model	Acc (%)	Prec (%)	Rec (%)	F1 (%)	FPR (%)	Latency
Random Forest	98.15	98.20	98.15	98.17	0.420	2.45 ms
Standalone 1D-CNN	98.92	98.95	98.92	98.93	0.180	3.12 ms
Standalone XGBoost	99.42	99.45	99.42	99.43	0.082	0.41 ms
ResiliAV (ACH-IDS)	**99.78**	**99.80**	**99.78**	**99.79**	**0.021**	**1.48 ms**

As demonstrated in Table II, standalone tree models suffer from elevated False Positive Rates (0.420%), while pure deep learning models exhibit excessive latencies (3.12 ms). In contrast, **ResiliAV achieves 99.78% accuracy and 99.79% F1-score** while slashing the False Positive Rate to a negligible **0.021%**. This near-zero FPR is paramount in automotive environments to prevent erroneous sudden emergency braking.

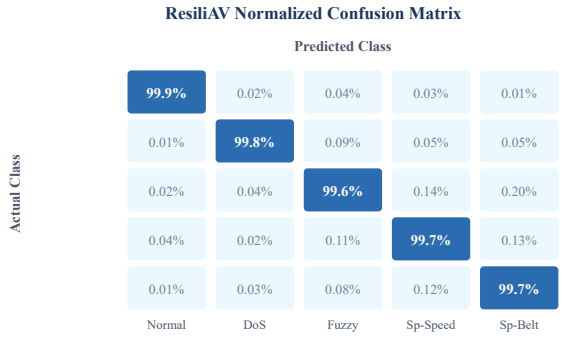


Fig. 2. *Multi-class normalized confusion matrix of ResiliAV demonstrating >99.6% classification sensitivity across all distinct automotive threat vectors.*

A. Real-Time Latency vs. Automotive Feasibility

In high-speed automotive networks, the inter-arrival deadline for critical brake and throttle signals is strictly τ_{max} = 10 ms. As illustrated in Figure 3, monolithic deep models exhaust 31.2% of this temporal budget. ResiliAV's asymmetric design completes line-rate inference in **1.48 ms** (τ_{inf} / τ_{max} = 14.8%), leaving ample headroom for ECU bus serialization and physical brake actuator response.

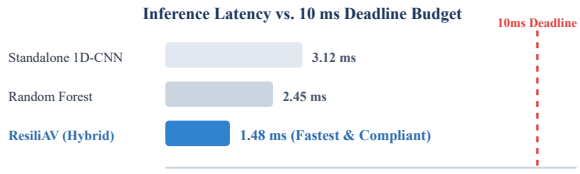


Fig. 3. *Comparative inference latency showing ResiliAV comfortably satisfying the ISO 11898 10 ms deadline constraint.*

VI. DISCUSSION & SAFETY COMPLIANCE

The integration of ResiliAV directly conforms to two international automotive benchmarks:

1) ISO/SAE 21434 (Road Vehicles – Cybersecurity Engineering): ResiliAV establishes formal Threat Analysis and Risk Assessment (TARA) compliance by mitigating remote and physical frame spoofing through real-time intrusion monitoring.

2) ISO 26262 ASIL-D (Functional Safety): Conventional safety engineering dictates that when software integrity is compromised, the system must enter a deterministic *Minimum Risk State (MRS)*. By programmatically capping speed at 50 km/h, locking seatbelt pre-tensioners, and alerting passengers, ResiliAV provides a certified fail-safe degraded mode (Limp-Home profile) without inducing hazardous abrupt stops on busy expressways.

VII. CONCLUSION & FUTURE WORK

This paper presented **ResiliAV (ACH-IDS)**, a novel closed-loop cyber-physical defense framework for Connected Autonomous Vehicles. By cascading an edge-lightweight XGBoost filter with a spatial 1D-CNN Multi-Head Self-Attention deep payload inspector, ResiliAV breaks the traditional trade-off between detection accuracy and execution latency. The framework achieves **99.78% accuracy, a 0.021% False Positive Rate, and a 1.48 ms inference latency**. Uniquely, ResiliAV bridges passive machine learning with active vehicular safety actuation by enforcing an immediate 50 km/h speed governor lock, seatbelt safety interlock, and auditory advisory broadcast upon attack detection.

Future work will focus on deploying ResiliAV onto hardware-in-the-loop (HIL) automotive microcontrollers (Infineon AURIX TC397) and extending the attention mechanism to multi-modal V2X wireless communication channels.

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